NUR E JANNAT ASHA

AI & Machine Learning Engineer | GenAI, LLMs & High-Performance Systems

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LinkedIn: linkedin.com/in/nur-e-jannat-asha | GitHub: github.com/Jannat332 | Portfolio: jannat332.github.io/portfolio

Work Authorization: Full Finnish Work Rights (LUT M.Sc. Graduate) | Immediate Availability | Aalto University F.E.C. Eligible

PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with a Master of Science in Software Engineering (**4.24/5.00 CGPA, Distinction**) from LUT University, Finland. Specialized in architecting reliable Generative AI pipelines, **Retrieval-Augmented Generation (RAG)**, explicit Knowledge Graph ontology validation, and GPU-accelerated computing (CUDA). Demonstrated success developing neuro-symbolic verification engines that reduce LLM hallucination rates by 42% on multi-hop benchmarks and accelerating computational modeling algorithms by 60% with CUDA kernels. Co-authored 5 peer-reviewed publications across IEEE and Elsevier journals. Accepted into the Aalto University F.E.C. recruitment program; available immediately for direct hire or subsidized placement.

TECHNICAL SKILLS

AI, ML & GenAI: Retrieval-Augmented Generation (RAG), Knowledge Graph Validation, Prompt Engineering, Hallucination Mitigation, Embeddings & Vector Search, Confidence Fusion Scoring

Languages & Frameworks: Python (3.10+, FastAPI, PyTorch, NumPy, Pandas, Scikit-Learn), C++17, CUDA, SQL (PostgreSQL, SQLite), Bash, React 18

High-Performance Computing: GPU Kernel Parallelization, Numerical Simulation, Matrix Computation, Cache Optimization, Asynchronous Processing

DevOps & MLOps: Docker (reproducible environments), GitHub Actions CI/CD, Linux/Unix, Git, RESTful API Design, Postman

Scientific & Quality: Mathematical Modeling, Statistical Error Analysis, Signal Acquisition & Noise Filtering (PPG / IoT Sensors), TDD (PyTest, GoogleTest)

Languages: English (Fluent / Full Professional Working Proficiency), Finnish (A1 - active weekly coursework), Bengali (Native)

TECHNICAL PROJECTS

LLM Parallel Reasoning & Verification Engine (Python 3.11, RAG, Knowledge Graph) github.com/Jannat332/llm-parallel-reasoning

Dual-Stream Semantic Retrieval & Dynamic Contradiction Scoring

- Designed a dual-stream reasoning architecture fusing dense vector retrieval with explicit knowledge-graph relational verification, **reducing factual hallucinations by 42%** on complex multi-hop queries.
- Implemented a confidence-scoring arbitration engine that quantifies semantic entropy and validates assertions before presenting generated output to users.
- Packaged benchmark evaluation suites and CLI runners inside containerized Docker images for deterministic, reproducible execution.

Quantitative Stock Signal & Technical Alert Engine (Python, Algorithms, Financial ML) github.com/Jannat332/stock-signal-alerts

Automated Market Momentum Indicators & Signal Trigger Pipeline

- Implemented mathematical technical indicator algorithms (SMA 20/50 Crossover, 14-period RSI, Momentum Divergence) with zero third-party financial calculation library dependency.
- Built an event-driven alert dispatcher generating machine-readable buy/sell signals with automated trend validation and threshold checking.
- Integrated comprehensive PyTest test suites verifying boundary conditions, zero-division safeguards, and extreme market volatility regimes.

Deterministic Embedded Traffic Controller (C++17, FSM, Safety Invariants) github.com/Jannat332/traffic-light-controller

Edge Computing & Safety-Critical Embedded State Machine

- Engineered deterministic state machine with thread-safe preemption and zero race conditions in modern C++17.
- Enforced 100% safety invariants with GoogleTest suite verifying hazard transitions and pedestrian walk timings.

Production CI/CD Matrix Pipeline Template (GitHub Actions, MLOps) github.com/Jannat332/github-actions-ci-template

Automated Testing, Multi-Stage Containerization & Quality Gatekeeping

- Constructed a GitHub Actions workflow automating linting, security vulnerability scanning, and multi-stage Docker builds with a **65% image footprint reduction**.

PROFESSIONAL EXPERIENCE

Research Assistant (Computational Systems) | North South University **2023 – 2024**

GPU Acceleration, Numerical Modeling & Simulation Workflows

- Accelerated scientific simulation algorithms by **60% using CUDA GPU kernel parallelization** and vectorization in C++ and Python.
- Engineered automated data preprocessing, numerical convergence checking, and statistical validation workflows for complex physical dynamics.
- Co-authored 3 computational research publications in prestigious international Elsevier scientific journals.

Junior Software Engineer | Spectrum IT Solutions **2021 – 2022**

Backend Engineering, Service Optimization & Linux Systems

- Developed and maintained production backend APIs and data endpoints in Python and C++ for enterprise clients.
- Conducted performance profiling and debugging with GDB and Valgrind, improving data query response latency by 25%.
- Collaborated in Agile sprints using Git workflows, automated test pipelines, and peer code reviews.

EDUCATION & QUALIFICATIONS

Master of Science in Software Engineering and Digital Transformation **2024 – 2025**

LUT University, Lappeenranta, Finland • ***CGPA: 4.24 / 5.00 (Distinction, Top 5%)***

- Curriculum: Advanced Software Architecture, Distributed Data Systems, Cloud Computing, Modern Engineering Practices.

Bachelor of Science in Computer Science and Engineering **2016 – 2021**

North South University • ***CGPA: 3.44 / 4.00***

- Embedded sensor system and physiological signal classification published in IEEE Xplore.

Aalto University F.E.C. (Further Educated with Companies) Program **Intake 2025 – 2026**

Aalto University, Espoo, Finland

- Accepted participant. Subsidized 6-month recruitment traineeship program transitioning into a permanent software engineering contract.

SELECTED SCIENTIFIC PUBLICATIONS

- **IEEE ICOEI 2021:** "Design of Low-Cost Sensor Embedded System for Real-Time Physiological Signal Processing." DOI: 10.1109/ICOEI51242.2021.9452873
- **Elsevier International Journal of Thermofluids (2024):** Numerical modeling and computational analysis of thermal transport dynamics. DOI: 10.1016/j.ijft.2024.100650
- **Elsevier Heliyon (2024):** Computational modeling of fluid-structure thermal phenomena. DOI: 10.1016/j.heliyon.2024.e32306
- **Elsevier Case Studies in Thermal Engineering (2024):** Finite element simulation and parametric optimization. DOI: 10.1016/j.csite.2024.104273